

MUGILAN C

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PROFESSIONAL SUMMARY

Passionate Computer Science graduate seeking a software engineering role where I can apply my full-stack development skills in Java, Python, React.js, Spring Boot, and Django to build impactful, real-world applications. Experienced in developing scalable web platforms, optimizing MySQL databases, and delivering measurable results through internships and academic projects. Eager to collaborate with a dynamic engineering team, contribute meaningful solutions, and grow as a developer in a challenging and fast-paced environment.

PROFESSIONAL EXPERIENCE

Software Trainee – Hindustan Aeronautics Limited (HAL)

Feb 2026 – Present | Bengaluru, KA

- Collaborate with the Backend and Database support team to maintain and optimize MySQL databases serving real-time aeronautics operations.
- Perform SQL query performance tuning, index optimization, and schema analysis, contributing to measurable improvements in query response times.
- Design and maintain stored procedures and backup strategies to ensure data integrity and high availability.
- Debug and resolve backend issues in real-time data processing workflows, reducing system downtime and improving reliability.
- Document database maintenance procedures and contribute to knowledge-sharing within the engineering team.

Full Stack Developer Intern – Extion Infotech

Oct 2024 – Feb 2025 | Bengaluru, KA

- Engineered core modules of a Blood Bank Management System, improving donor registration speed by 30% through optimized form flows and API integration.
- Built responsive, user-facing pages including donor sign-up and admin login dashboards using React.js with component-based architecture.
- Implemented role-based access control (RBAC), JWT authentication, and input validation to enforce data security across the platform.
- Collaborated in an Agile team to debug UI/UX issues, perform code reviews, and ship iterative feature releases.

PROJECTS

Guild-Based Online Coding Platform | React.js, Spring Boot, MySQL, Docker, WebSockets [GitHub](#)

- Architected an AI-driven coding platform featuring guild challenges, leaderboards, and real-time collaboration, increasing learner engagement by 30%.
- Containerized automated code evaluation using Docker, executing test cases in isolated environments and reducing incorrect evaluations by 20%.
- Implemented real-time multiplayer collaboration using WebSockets (Socket.io), enabling live co-coding and instant leaderboard updates.
- Published findings as a peer-reviewed research paper: "Guild-Based Online Coding Platform and Gamification Features" — IEEE Xplore.

Bike Rental System | Next.js, Django, HTML, CSS, JavaScript, MySQL [GitHub](#)

- Developed a full-stack bike rental web application with real-time availability tracking, online booking, and dynamic pricing logic.
- Built RESTful APIs with Django REST Framework, integrating seamlessly with a Next.js frontend for efficient server-side rendering and data fetching.
- Designed normalized relational database schemas in MySQL to support concurrent bookings, rental history, and dynamic pricing rules.
- Delivered fully responsive UI across mobile and desktop, ensuring consistent user experience and optimized page load performance.

EDUCATION

CMR Institute of Technology

B.E. in Computer Science | CGPA: 7.99 / 10.0

Dec 2021 – Jun 2025

CMR National PU College

Higher Secondary – Computer Science | Score: 80.66%

Jul 2019 – Jul 2021

Poornaprajna Education Centre

SSLC (Class 10) | Score: 82.08%

May 2018 – Apr 2019

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL

Frontend: React.js, Next.js, HTML5, CSS3

Backend: Spring Boot, Django, Django REST Framework, RESTful APIs

Databases: MySQL, Query Optimization, Stored Procedures, Indexing

ACHIEVEMENTS & CERTIFICATIONS

- IEEE Xplore Publication: "Guild-Based Online Coding Platform and Gamification Features" — peer-reviewed and indexed in IEEE digital library.
- Java for Beginners – Comprehensive Java fundamentals course covering OOP, data structures, and core Java APIs.
- 100 Days of Python Bootcamp – Completed Udemy's intensive Python bootcamp covering automation, data handling, and web development.